

PUBLICATIONS AND DOCUMENTED RESEARCH ACTIVITIES

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Peer-reviewed journal articles

Saber, M., Posite, V. R., Koroma, A. O., Ly, A., Tejan Bah, A. M., Abdulai, A., et al. (2026). Climate change hotspots in Africa: spatiotemporal trends of rainfall, temperature, and drought using GIS and statistical modelling. *Journal of Earth System Science*, 135, Article 187.

Continental-scale empirical study identifying climate hotspots across Africa. Contribution centred on construction, validation and administrative-unit aggregation of gridded climate datasets, statistical trend detection across a multi-decade panel, and GIS-based spatial modelling of hazard exposure.

Saber, M., Koroma, A., Posite, R. V., Ly, A., Tejan Bah, A. M., Abdulai, A., et al. (2025). A comprehensive review of climate change adaptation and disaster risk reduction in Africa. *Journal of Water and Climate Change*, 16(5), 1831–1862.

Systematic review synthesising adaptation and disaster risk reduction literature across Africa, covering strategies, institutional frameworks, financing mechanisms and the evidence base on effectiveness.

Book chapter

Siddique, H. A., Kanagie, A., and Abdulai, A. (2024). Echoes of the past: Power and colonial impact on contemporary climate politics. In *Climate change and environmental politics* (pp. 141–166). Transnational Press London.

Analytical chapter on how colonial-era power structures continue to shape the distribution of responsibility, the terms of climate finance, and asymmetries in negotiating position in international climate politics.

Master's theses

Abdulai, A. (2026). Path Dependency and Cement Industry Decarbonisation in the UK: A Case Study of the Peak Cluster CCS Project. MSc thesis, 30 ECTS, Grade A. Central European University — MESPOM Erasmus Mundus. Supervisor: Prof. Ian Kane, University of Manchester.

Modelled abatement costs by Monte Carlo simulation under uncertainty in capital expenditure, operating cost, capture rate and carbon price, benchmarking resulting cost distributions against UK Emissions Trading Scheme allowance prices to identify conditions under which capture investment becomes economically rational. Analysed technological and institutional lock-in in a hard-to-abate sector through the framework of path dependency. Findings indicated that viability depends more on the credibility and trajectory of the carbon price, and on coordination across a multi-firm cluster sharing transport and storage infrastructure, than on the central technology cost estimate.

Abdulai, A. (2024). The role of blockchain in establishing transparent carbon markets in Africa. MSc thesis, 18 ECTS, 18.55/20. Pan African University Institute of Water and Energy Sciences (PAUWES), Université Abou Bekr Belkaid, Tlemcen.

Combined three methods: a Decision Framework Evaluation developing and validating structured decision criteria for project developers; DEMATEL (Decision Making Trial and Evaluation Laboratory) multi-criteria decision analysis implemented in MATLAB, converting expert judgements into a direct-relation matrix, normalising it, computing total-relation effects and clustering variables into cause and effect groups; and design of a conceptual carbon trading architecture aligned to the UNFCCC monitoring and reporting framework for Article 6.4 ITMO transfer and NDC tracking. Identified smart contracts, data security, regulatory frameworks and transparency as dominant and interdependent adoption enablers.

Undergraduate dissertation

Abdulai, A. (2019). The Effects of Climate Change and Variability on Irrigated Agriculture in Northern Ghana. BSc dissertation, University for Development Studies, Ghana.

Conference presentations and posters

Mapping Desertification in Northern Ghana using Machine Learning. Poster, Advanced Climate Change Conference, PAUWES, Tlemcen. Applied Normalised Difference Vegetation Index and Enhanced Normalised Difference Impervious Surface Index methods to Sentinel-2 and Landsat 8 imagery to compare vegetative cover across baseline and post-baseline periods across five regions of Northern Ghana.

Community-Based Flood Risk Management in Algeria: A Critical Review of Current Issues and Challenges. Poster, International Symposium on Flash Floods, Algeria.

Socioeconomic Vulnerability and Adaptation Measures: A Review of Flash Flood Impacts under Changing Climate in Algeria. Poster, International Symposium on Flash Floods, Algeria.

The Youth and Climate Action: Towards Realizing the Africa We Want. Invited lightning talk, Advanced Climate Science Conference, PAUWES, Tlemcen (2022).

Applied research and analytical outputs

Africa Carbon Markets Initiative, SEforALL (2024). Research on validation and verification body capacity across Africa; quantitative and qualitative analysis of carbon market activity informing private-sector development;

methodology development research including diesel displacement. Core team member on development and launch of a digital platform tracking sustainability metrics for carbon credit projects in Nigeria.

UNFCCC Secretariat, Transparency for Developing Countries Division (2024). Review and analysis of Biennial Update Reports submitted by developing countries, identifying gaps and inconsistencies in reporting methodology and data quality; management of greenhouse gas inventory and emission reduction target data.

UNFCCC Regional Collaboration Centre, East and Southern Africa (2023–24). Research on Article 6.4 project cycles and carbon credit accounting; structured interviews with carbon project developers and analysis of resulting data for regional reporting.

Common Fund for Commodities and Forestry and Land Check (2025). Synthesis of 20+ technical reports into structured analytical briefs and KPI documents; quality assurance of 15+ programme documents against external funder methodological standards.

Innovations for Poverty Action, Escaping Poverty Project (2019–21). Multi-year randomised controlled trial. Field survey data monitoring, discrepancy reconciliation and high-frequency checks; supervision of 2 data auditors and 10 field agents; cleaning and analysis of quantitative panel datasets in Stata.

Youth Challenge International, Ghana (2021–22). Designed an environmental risk assessment tool enabling social entrepreneurs to map emissions and sustainability exposure; secondary research on green business, social entrepreneurship and country context to inform programme design.

Research awards and competitive distinctions

- Schneider Electric Go Green Global Challenge 2025 – Winner (6000 EUR Prize Money)
- Stanford Doerr Global Sustainability Challenge 2026 – Top Six, Won Jury Prize for Strategic Thinking
- Geneva Challenge 2023 — 2nd place, Global Contest, Geneva; 1st place, Africa.
- EU Erasmus Mundus Joint Master's Scholarship (2024–2026).
- African Union scholarship, Pan African University (2022–2024).
- Best Graduating Student, Agribusiness — Climate Change and Food Security, University for Development Studies (2019).

Research training

- Certificate in Climate Diplomacy — Konrad-Adenauer-Stiftung, Regional Programme on Energy Security and Climate Change, Kenya (Apr–Jun 2023).
- Energy scenario modelling using OSeMOSYS.
- Digital Earth Africa Sandbox for Earth observation and analysis.